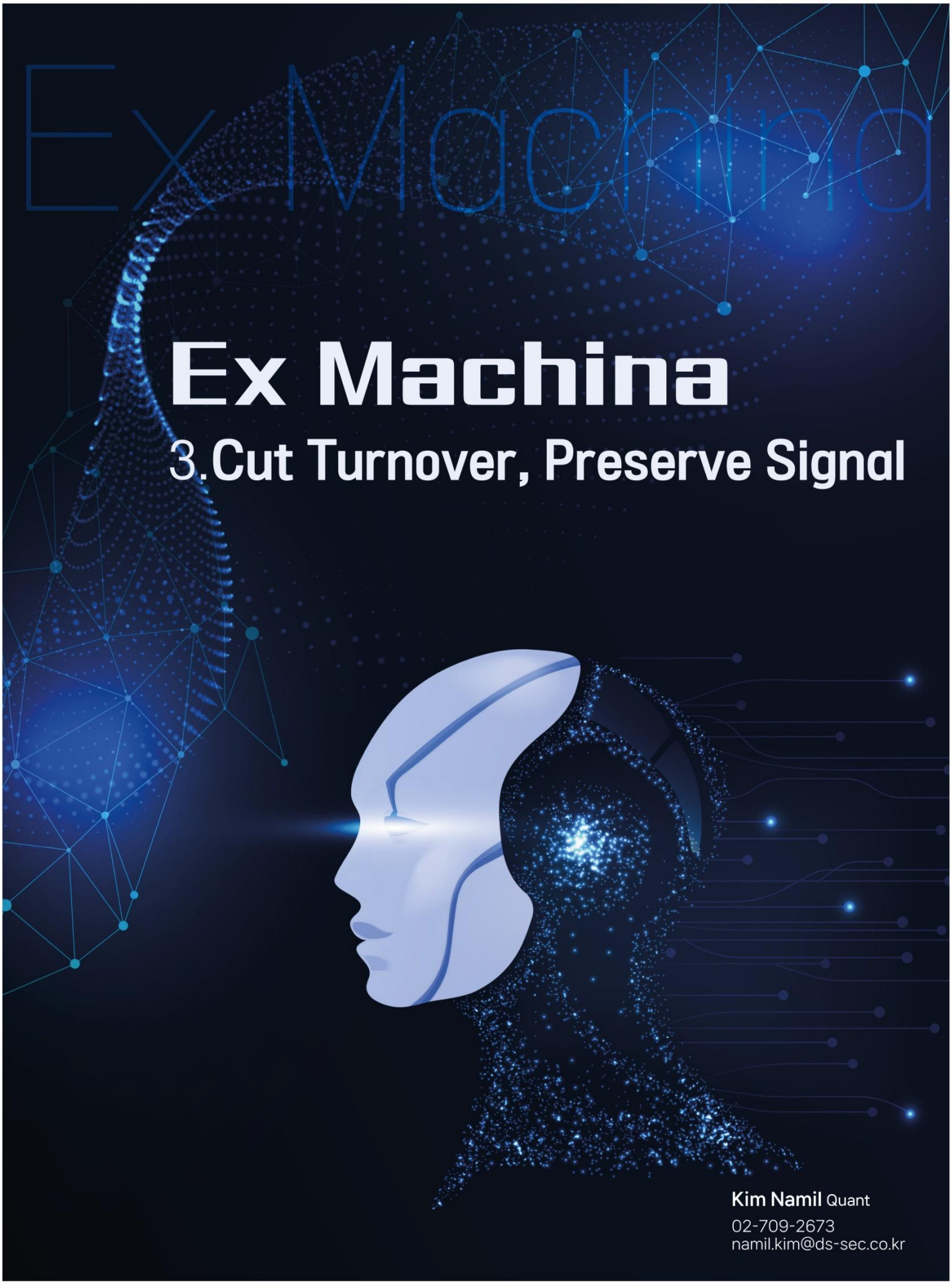
DS INVESTMENT & SECURITIES

QUANT



Ex Machina

3.Cut Turnover, Preserve Signal

Kim Namil Quant
02-709-2673
namil.kim@ds-sec.co.kr

Cut Turnover, Preserve Signal

1. Setup and Conservation of Total Weight

Total weight conserved before costs

Let B be portfolio market value and $\boldsymbol{w}^-$ current long-only weights. Fix m divisible buy-sell pairs, jointly feasible in full, with amounts $a_j > 0$. Define $b_j = a_j/B$, $\boldsymbol{b} = (b_1, \dots, b_m)^\top$, and $T = \sum_j b_j > 0$. Let $\boldsymbol{e}$ denote a standard basis vector, $C_j = \boldsymbol{e}_{\text{buy},j} - \boldsymbol{e}_{\text{sell},j}$, and $C = [C_1 \ \cdots \ C_m]$. Before costs, the weight sum is conserved (D_q is defined in (2)):

$$C_{\text{ex}} = \begin{pmatrix} -1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \boldsymbol{w}(q) = \boldsymbol{w}^- + CD_q\boldsymbol{b}, \qquad \mathbf{1}^\top C = \mathbf{0}^\top. \tag{1}$$

2. Score-Prioritized Rebalancing Rule

Descending-gain execution; only the boundary pair is partial

Let s be the asset score. Index pairs $j = 1, \dots, m$ in descending order of $g_j = s_{\text{buy},j} - s_{\text{sell},j} > 0$. Execute a fraction $q \in [0, T]$ of B; only the boundary pair may be partially filled. The execution-fraction matrix is

$$D_q = \operatorname{diag}(\theta_1, \dots, \theta_m), \qquad \theta_j = \min\left\{1, \max\left\{0, \frac{q - \sum_{\ell<j} b_\ell}{b_j}\right\}\right\}. \tag{2}$$

For disjoint buy and sell sets, one-way turnover [1] is

$$\text{TO}(q) = \frac{1}{2}\left\|CD_q\boldsymbol{b}\right\|_1 = \sum\nolimits_j \theta_j b_j = q, \qquad r := \frac{\text{TO}(q)}{T} = \frac{q}{T}. \tag{3}$$

Optimality rationale. Within the fixed feasible pair set, shifting a feasible weight $\delta > 0$ from pair j to pair $i < j$ adds $\delta(g_i - g_j) \geq 0$. Thus descending execution maximizes $\sum_j g_j \theta_j b_j$ at fixed q.

3. Signal Preservation and Net Performance

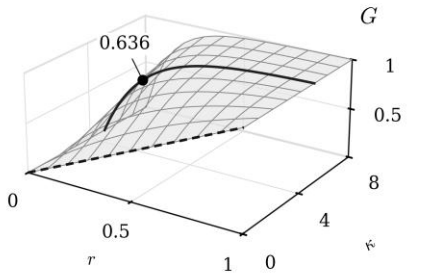
At r = 0.2: 80% less turnover, 63.6% of improvement kept

Lay out g_j as a step function on the cumulative traded-value fraction $u \in [0,1]$, assigning width b_j/T to pair j. Approximate it by $\gamma(u) \propto e^{-\kappa u}$ with the same integral, so $H = \sum_j b_j g_j = T\int_0^1 \gamma(u)\,\mathrm{d}u$. For signal concentration $\kappa > 0$ and execution fraction $0 \leq r \leq 1$:

$$G(r,\kappa) = \frac{\int_0^r \gamma(u)\,\mathrm{d}u}{\int_0^1 \gamma(u)\,\mathrm{d}u} = \frac{1 - e^{-\kappa r}}{1 - e^{-\kappa}} \geq r. \tag{4}$$

As $\kappa \to 0$, $G = r$. For $T = 0.5$, $q = 0.1$, and $\kappa = 5, r = 0.2$: 80% less traded value (turnover) and 63.6% of modeled score improvement captured.

Figure 1. Captured score improvement G(r, κ)



Source: DS Investment & Securities Research Center
Note: Solid: $\kappa = 5$; dashed: $\kappa = 0$, $G = r$.

If scores represent expected returns, H is full expected gain. Let $c \geq 0$ be a uniform cost rate per replacement amount, combining buying and selling costs. Cost relative to B is $c \cdot \text{TO}(q) = c \cdot T \cdot r$:

$$N(r) := HG - c \cdot T \cdot r, \qquad N(r) > N(1) \iff cT(1-r) > H(1-G). \tag{5}$$

Break-even. In this example, $N(0.2) > N(1)$ requires $cT > 0.45449$. For $H, T, c, \kappa > 0$, the concave model gives

$$r^* = \left[\frac{1}{\kappa}\ln\left(\frac{H\kappa}{cT(1-e^{-\kappa})}\right)\right]_0^1, \qquad q^* = Tr^*. \tag{6}$$

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